SUPERSEDING
MIL-W-13154(SigC)
8 December 1953

MILITARY SPECIFICATION

WIRE, ELECTRICAL; ANTENNA, HIGH FREQUENCY,
(WIRE W-146)

1. SCOPE

1.1 This specification covers a single conductor No. 12 AWG high-frequency antenna lead wire, polyethylene insulated, braided, asphalt impregnated.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards and publications of the issue in effect on the date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| J-C-98 | Cable and Wire, Insulated; Methods of Sampling and Testing. |
| L-P-390 | Plastic, Molding Material, Polyethylene, Low and Medium Density. |
| QQ-W-343 | Wire, Electrical and Nonelectrical, Copper (Uninsulated). |

MILITARY

| | |
|---|---|
| MIL-I-3042 | Insulation, Electrical, Cotton-Fiber, Untreated. |
| MIL-C-12000 | Cable, Cord, and Wire, Electric; Packaging of |

FSC 6145

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STANDARDS

MILITARY

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing Materials (ASTM)

| | |
|---|---|
| D36 | Test for Softening Point of Bituminous Materials (Ring-and-Ball Method). |
| D5 | Test for Penetration of Bituminous Materials. |
| D6 | Test for Loss on Heating of Oil and Asphaltic Compounds. |

(Copies of ASTM specifications may be obtained from the American Society for Testing Materials, 1916 Race St., Philadelphia, Pa., 19103)

3. REQUIREMENTS

3.1 Materials.

3.1.1 Strands.- Strands shall be 0.0185 inch, soft or drawn and annealed tinned copper wire.

3.1.2 Insulation.- Insulation compound shall be a polyethylene compound meeting the requirements of L-P-390, Type II, Class L, Grade 4.

3.1.3 Braid.- The braid shall be composed of 20/2 cotton, 3 ends, 18 picks, 6 carriers, and meet the performance requirements of MIL-I-3042 as applicable.

3.1.3.1 Braid saturant.- The conductor outer braid shall be thoroughly saturated with an asphaltic-base saturant. The saturant used shall be an asphaltic material blended with suitable high melting-point waxes, but no volatile material such as paraffin, scale wax, wax tailings, or flux oil shall be added. The saturant shall satisfy the following requirements of 3.1.3.1.1 through 3.1.3.1.4.2 inclusive.



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3.1.3.1.1 The softening point shall be not less than 140°F when determined in accordance with the method outlined in the American Society for Testing Materials Standard Method of Test for Softening Point of Bituminous Materials (Ring-and-Ball Method) - Serial Designation: D36.

3.1.3.1.2 The penetration at 77°F shall be between 22 and 32 when determined in accordance with the method outlined in the American Society for Testing Materials Standard Method of Test for Penetration of Bituminous Materials - Serial Designation: D5.

3.1.3.1.3 The loss of weight on heating shall be not more than 0.3 percent of the original weight when determined in accordance with the method outlined in the American Society for Testing Materials, Standard Method of Test for Loss on Heating of Oil and Asphaltic Compounds - Serial Designation: D6.

3.1.3.1.4 Retest after loss on heating.- After the loss of weight on heating test (paragraph 3.1.3.1.3), the saturant used in this test (after crust has been removed) shall be retested and shall then meet the following requirements:

3.1.3.1.4.1 The maximum rise in the softening point of the saturant over the original value shall not exceed 10°F.

3.1.3.1.4.2 The maximum lowering of the penetration value from the original value shall not exceed 10 and in no case shall the penetration be lower than 15.

3.1.3.1.5 Saturation of the braid covering with the saturant specified shall be so thorough that no white or unsaturated ends will be exposed by a diagonal cut through the covering.

3.1.3.2 Finishing compound.- The saturated braid shall be smoothly covered with a continuous layer of finishing compound which will adhere firmly to the braid. The finishing compound shall conform with the following and shall provide a hard, polished finish.

3.1.3.2.1 The melting point shall not be less than 180°F when determined in accordance with the method outlined in the American Society for Testing Materials Standard Method of Test for Softening Point of Bituminous Materials (Ring-and-Ball Method) - Serial Designation: D36.

3.1.3.2.2 The penetration of 77°F shall not exceed 9 when determined in accordance with the method outlined in the American Society for Testing Materials Standard Method of Test for Penetration of Bituminous Materials - Serial Designation: D5.

3.1.3.2.3 The loss of weight on heating shall be not more than 0.2 percent of the original weight when determined in accordance with the method outlined in the American Society for Testing Materials Standard Method of Test for Loss on Heating of Oil and Asphaltic Compounds - Serial Designation: D6.

3.1.3.3 Braid-low temperature.- The saturated braid shall not crack when tested in accordance with 4.3.



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3.2 Construction

3.2.1 Conductors.- Conductors shall be equivalent to 6530 circular mils (NO. 12 AWG) and shall consist of 19 strands of wire specified in 3.1.1. The conductors shall be continuous and shall conform to the requirements of Specification QQ-W-343 for Type C, Class C wire.

3.2.2 Insulation.- The insulation over the conductor shall have a minimum diameter of 0.357 inch, the nominal 0.370 inch, and the maximum 0.383 inch. The minimum insulation thickness shall be not less than 85 percent of the average thickness.

3.2.3 Braid.- A single braid shall be applied over the insulated conductor and saturated with asphalt compound, in conformance with 3.1.3.

3.2.4 Overall diameter.- The overall diameter of wire W-146 shall be nominal 0.420 inch, maximum 0.430 inch.

3.3 Physical properties.

3.3.1 Dielectric strength.- The finished wire shall be capable of withstanding a voltage of 12,500 volts rms, 60 cycle a.c. for one minute between conductor and water after immersion for 12 hours.

3.3.2 Insulation resistance.- After a potential of not less than 125 volts has been applied for one minute on the finished wire, the insulation resistance shall be not less than 10,000 megohms - 1000 feet.

3.3.3 Dielectric constant.- The dielectric constant of the finished wire shall not exceed 4, when measured in mercury at 1000 c.p.s. at or corrected to a temperature of + 25° C.

3.3.4 Cold bend.- There shall be no evidence of cracking of the insulation when the wire is given the cold bend test of 4.10.

3.4 Workmanship.- All wire shall be manufactured in a thoroughly workmanlike manner.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- Inspection shall be classified as follows:

(a) Inspection covered by subsidiary documents (see 4.3)
(b) Quality conformance inspection before preparation for delivery (see 4.4)
(c) Quality conformance inspection of preparation for delivery (see 4.11).



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4.3 Inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the quality conformance inspection before preparation for delivery:

| Item | When Required |
|---|---|
| Conductors | 3.2.1 |
| Insulation | 3.1.2 |
| Braid | 3.1.3 |

4.4 Quality conformance inspection of preparation for delivery.- The supplier shall perform the inspection specified in 4.3 and 4.4.1 through 4.4.2. This does not relieve the supplier of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the supplier's inspection procedures and examine the supplier's inspection records. In addition, the Government - at its discretion - may perform all or any part of the specified inspection to verify the supplier's compliance with specification requirements. Test equipment for Government verification inspection shall be made available by the supplier.

4.4.1 Group A inspection.- This inspection, including sampling, shall conform to Table I and the ordinary inspection procedures of Standard MIL-STD-105.

Table I. Group A Inspection

| Inspection | Req. Para. | Insp. Para. | AQL | |
|---|---|---|---|---|
| | | | Major | Minor |
| Visual & Dimensional | | | | |
| Conductors | 3.2.1 | 4.5 | 1.0% for the group | 4.0% for the group |
| Insulation | 3.2.2 | 4.5 | | |
| Braid | 3.2.3 | 4.5 | | |
| Overall diameter | 3.2.4 | 4.5 | | |
| Electrical | | | | |
| Electrical continuity | 3.2.1 | 4.6 | 1.0% for the group | ---- |
| Dielectric strength | 3.3.1 | 4.7 | | |
| Insulation resistance | 3.3.2 | 4.8 | | |



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4.4.2 Group B inspection.- The inspection, including sampling, shall conform to Table II and to special sampling level S-4 of MIL-STD-105 for normal, tightened and reduced inspection. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on sample units that have been subjected to and met Group A inspection.

Table II. Group B Inspection

| Inspection | Req. Para. | Insp. Para. | AQL Major | AQL Minor |
|---|---|---|---|---|
| Dielectric Constant | 3.3.3 | 4.9 | 4.0% combined | --- |
| Cold Bend | 3.3.4 | 4.10 | | --- |

4.5 Visual and dimensional.- Wire W-146 shall be inspected to verify that the material and workmanship comply with this specification.

4.6 Continuity.- The wire shall be tested for continuity using a potential of not more than 10 volts. A suitable indicator such as a flashlight bulb shall be placed in the circuit. This test shall be performed prior to the dielectric strength test of 4.7.

4.7 Dielectric strength test.- Wire W-146 shall be tested in accordance with method 6111 of Specification J-C-98 to determine compliance with the requirements of 3.3.1.

4.8 Insulation resistance.- Method 6031 of Specification J-C-98 shall be used to determine compliance with the requirements of 3.3.2.

4.9 Dielectric constant.- Dielectric constant shall be measured with a capacitance bridge. The bridge shall be capable of measuring the capacitance within an accuracy of 1 mmf. It shall also be capable of measuring the capacitance of a specimen, one side of which is grounded. The wire shall meet the requirements of 3.3.3.

4.10 Cold bend.- The samples of Wire W-146 shall be attached to a 2 inch diameter mandrel and suspended vertically with lower ends weighted sufficiently to keep specimens taut and to permit bending them without handling. The mandrels and specimens shall be placed for at least 20 hours in a cold chamber at -40° C and while at this temperature bent for 5 close turns around the mandrel at the rate of approximately 15 turns per minute. After completion of the test the insulation shall be examined through a magnifying glass of at least 3 diameters magnification.

4.11 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected to determine compliance with the requirements of Section 5.



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5. PREPARATION FOR DELIVERY

5.1 Wires shall be preserved, packaged, packed and marked in accordance with Specification MIL-C-12000. (See 6.2).

5.2 Preservation, packaging and packing shall be level A, B, or C as specified. (See 6.2).

6. NOTES

6.1 Intended use.- Used as antenna lead wire.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Wire type designation.
(c) Length of wire required.
(d) Level of packaging and level of packing required. (See 5.1, 5.2)
(e) Place of final inspection.

6.3 Definitions

6.3.1 Inspection lot.- An inspection lot shall consist of all units of product produced under essentially the same conditions and submitted for acceptance at the same time.

6.3.2 Unit of product.- A unit of product is the continuous length of wire contained on a single reel or spool or in a package.

6.3.3 Specimen.- A specimen consists of the individual piece of the unit of product upon which the test is made.

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Project No. 6145-A038



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# SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

**INSTRUCTIONS**

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

| SPECIFICATION | | |
|---|---|---|
| ORGANIZATION | CITY AND STATE | |
| CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $ |

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

| SUBMITTED BY *(Printed or typed name and activity)* | DATE |
|---|---|
| | |

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